

# DATA DICTIONARY

TABLE 1: Role

| Attribute   | Data Type                    | Constraints                 | Description                                |
|-------------|------------------------------|-----------------------------|--------------------------------------------|
| RoleID      | INT                          | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for each role            |
| RoleName    | VARCHAR(50) NOT NULL, UNIQUE |                             | Name of role (Guest, Host, Admin, Support) |
| Permissions | JSON                         |                             | JSON structure containing role permissions |
| CreatedAt   | TIMESTAMP                    | DEFAULT CURRENT_TIMESTAMP   | Timestamp when role was created            |

TABLE 2: User

| Attribute      | Data Type                     | Constraints                         | Description                        |
|----------------|-------------------------------|-------------------------------------|------------------------------------|
| UserID         | INT                           | PRIMARY KEY, AUTO_INCREMENT         | Unique identifier for each user    |
| RoleID         | INT                           | FOREIGN KEY (Role), NOT NULL        | References Role table              |
| FirstName      | VARCHAR(50)                   | NOT NULL                            | User's first name                  |
| LastName       | VARCHAR(50)                   | NOT NULL                            | User's last name                   |
| Email          | VARCHAR(100) NOT NULL, UNIQUE |                                     | User's email address               |
| Phone          | VARCHAR(20)                   |                                     | Contact phone number               |
| ProfilePic     | VARCHAR(255)                  |                                     | URL to profile picture             |
| PasswordHash   | VARCHAR(255) NOT NULL         |                                     | Hashed password for authentication |
| JoinDate       | DATE                          | DEFAULT CURRENT_DATE                | Date user joined the platform      |
| VerifiedStatus | BOOLEAN                       | DEFAULT FALSE                       | Whether user identity is verified  |
| LastLogin      | TIMESTAMP                     | NULL                                | Last login timestamp               |
| CreatedAt      | TIMESTAMP                     | DEFAULT CURRENT_TIMESTAMP           | Record creation timestamp          |
| UpdatedAt      | TIMESTAMP                     | DEFAULT CURRENT_TIMESTAMP ON UPDATE | Last update timestamp              |

**TABLE 3: Location**

| Attribute  | Data Type     | Constraints                 | Description                         |
|------------|---------------|-----------------------------|-------------------------------------|
| LocationID | INT           | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for each location |
| Address    | VARCHAR(255)  | NOT NULL                    | Street address                      |
| City       | VARCHAR(100)  | NOT NULL                    | City name                           |
| State      | VARCHAR(100)  |                             | State or province (nullable)        |
| Country    | VARCHAR(100)  | NOT NULL                    | Country name                        |
| PostalCode | VARCHAR(20)   |                             | Postal or ZIP code                  |
| Latitude   | DECIMAL(10,8) |                             | Geographic latitude coordinate      |
| Longitude  | DECIMAL(11,8) |                             | Geographic longitude coordinate     |
| CreatedAt  | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP   | Record creation timestamp           |

**TABLE 4: PropertyType**

| Attribute   | Data Type    | Constraints                 | Description                                    |
|-------------|--------------|-----------------------------|------------------------------------------------|
| TypeID      | INT          | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for property type            |
| TypeName    | VARCHAR(100) | NOT NULL, UNIQUE            | Name of property type (Apartment, House, etc.) |
| Description | TEXT         |                             | Detailed description of property type          |
| CreatedAt   | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP   | Record creation timestamp                      |

**TABLE 5: Property**

| Attribute  | Data Type    | Constraints                          | Description                         |
|------------|--------------|--------------------------------------|-------------------------------------|
| PropertyID | INT          | PRIMARY KEY, AUTO_INCREMENT          | Unique identifier for each property |
| HostID     | INT          | FOREIGN KEY (User), NOT NULL         | References User table (host)        |
| TypeID     | INT          | FOREIGN KEY (PropertyType), NOT NULL | References PropertyType table       |
| Title      | VARCHAR(200) | NOT NULL                             | Property listing title              |

| Attribute    | Data Type     | Constraints                         | Description                         |
|--------------|---------------|-------------------------------------|-------------------------------------|
| Description  | TEXT          |                                     | Detailed property description       |
| LocationID   | INT           | FOREIGN KEY (Location), NOT NULL    | References Location table           |
| BasePrice    | DECIMAL(10,2) | NOT NULL, CHECK > 0                 | Standard price per night            |
| CleaningFee  | DECIMAL(8,2)  | DEFAULT 0                           | Additional cleaning fee             |
| ServiceFee   | DECIMAL(8,2)  | DEFAULT 0                           | Platform service fee                |
| MaxGuests    | INT           | NOT NULL, CHECK > 0                 | Maximum number of guests allowed    |
| Bedrooms     | INT           |                                     | Number of bedrooms                  |
| Bathrooms    | DECIMAL(3,1)  |                                     | Number of bathrooms                 |
| SquareMeters | INT           |                                     | Property size in square meters      |
| IsActive     | BOOLEAN       | DEFAULT TRUE                        | Whether listing is currently active |
| CreatedAt    | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP           | Record creation timestamp           |
| UpdatedAt    | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP ON UPDATE | Last update timestamp               |

**TABLE 6: Amenity**

| Attribute   | Data Type    | Constraints                 | Description                                     |
|-------------|--------------|-----------------------------|-------------------------------------------------|
| AmenityID   | INT          | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for each amenity              |
| AmenityName | VARCHAR(100) | NOT NULL, UNIQUE            | Name of amenity (WiFi, Pool, etc.)              |
| Category    | VARCHAR(50)  |                             | Category of amenity (Essentials, Outdoor, etc.) |
| IconClass   | VARCHAR(50)  |                             | CSS class for icon                              |
| CreatedAt   | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP   | Record creation timestamp                       |

**TABLE 7: PropertyAmenity (Bridge Table)**

| Attribute  | Data Type | Constraints                         | Description                        |
|------------|-----------|-------------------------------------|------------------------------------|
| PropertyID | INT       | PRIMARY KEY, FOREIGN KEY (Property) | References Property table          |
| AmenityID  | INT       | PRIMARY KEY, FOREIGN KEY (Amenity)  | References Amenity table           |
| AddedDate  | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP           | When amenity was added to property |

**TABLE 8: BookingStatus**

| Attribute   | Data Type    | Constraints                 | Description                               |
|-------------|--------------|-----------------------------|-------------------------------------------|
| StatusID    | INT          | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for booking status      |
| StatusName  | VARCHAR(50)  | NOT NULL, UNIQUE            | Name of status (Pending, Confirmed, etc.) |
| Description | VARCHAR(255) |                             | Description of what status means          |
| IsActive    | BOOLEAN      | DEFAULT TRUE                | Whether this status is currently in use   |
| CreatedAt   | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP   | Record creation timestamp                 |

**TABLE 9: Booking**

| Attribute    | Data Type | Constraints                      | Description                        |
|--------------|-----------|----------------------------------|------------------------------------|
| BookingID    | INT       | PRIMARY KEY, AUTO_INCREMENT      | Unique identifier for each booking |
| GuestID      | INT       | FOREIGN KEY (User), NOT NULL     | References User table (guest)      |
| PropertyID   | INT       | FOREIGN KEY (Property), NOT NULL | References Property table          |
| CheckInDate  | DATE      | NOT NULL                         | Start date of stay                 |
| CheckOutDate | DATE      | NOT NULL, CHECK > CheckInDate    | End date of stay                   |
| Nights       | INT       | GENERATED (DATEDIFF)             | Calculated stay duration           |
| GuestsCount  | INT       | NOT NULL, DEFAULT 1, CHECK > 0   | Number of guests for this booking  |

| Attribute       | Data Type     | Constraints                           | Description                     |
|-----------------|---------------|---------------------------------------|---------------------------------|
| BasePrice       | DECIMAL(10,2) | NOT NULL                              | Base price at time of booking   |
| CleaningFee     | DECIMAL(8,2)  | DEFAULT 0                             | Cleaning fee at time of booking |
| ServiceFee      | DECIMAL(8,2)  | DEFAULT 0                             | Service fee at time of booking  |
| TotalPrice      | DECIMAL(12,2) | NOT NULL                              | Total price for the entire stay |
| StatusID        | INT           | FOREIGN KEY (BookingStatus), NOT NULL | References BookingStatus table  |
| SpecialRequests | TEXT          |                                       | Any special requests from guest |
| CreatedAt       | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP             | Record creation timestamp       |
| UpdatedAt       | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP ON UPDATE   | Last update timestamp           |

**TABLE 10: PaymentMethod**

| Attribute  | Data Type   | Constraints                 | Description                                |
|------------|-------------|-----------------------------|--------------------------------------------|
| MethodID   | INT         | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for payment method       |
| MethodName | VARCHAR(50) | NOT NULL, UNIQUE            | Name of method (Credit Card, PayPal, etc.) |
| Processor  | VARCHAR(50) |                             | Payment processor (Stripe, PayPal, etc.)   |
| IsActive   | BOOLEAN     | DEFAULT TRUE                | Whether method is currently available      |
| CreatedAt  | TIMESTAMP   | DEFAULT CURRENT_TIMESTAMP   | Record creation timestamp                  |

**TABLE 11: PaymentStatus**

| Attribute       | Data Type | Constraints                 | Description                          |
|-----------------|-----------|-----------------------------|--------------------------------------|
| PaymentStatusID | INT       | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for payment status |

| Attribute   | Data Type    | Constraints               | Description                               |
|-------------|--------------|---------------------------|-------------------------------------------|
| StatusName  | VARCHAR(50)  | NOT NULL, UNIQUE          | Name of status (Pending, Completed, etc.) |
| Description | VARCHAR(255) |                           | Description of status                     |
| CreatedAt   | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP | Record creation timestamp                 |

**TABLE 12: Payment**

| Attribute     | Data Type     | Constraints                           | Description                        |
|---------------|---------------|---------------------------------------|------------------------------------|
| PaymentID     | INT           | PRIMARY KEY, AUTO_INCREMENT           | Unique identifier for each payment |
| BookingID     | INT           | FOREIGN KEY (Booking), NOT NULL       | References Booking table           |
| Amount        | DECIMAL(10,2) | NOT NULL, CHECK > 0                   | Payment amount                     |
| PaymentDate   | DATETIME      | NOT NULL                              | Date and time of payment           |
| MethodID      | INT           | FOREIGN KEY (PaymentMethod), NOT NULL | References PaymentMethod table     |
| StatusID      | INT           | FOREIGN KEY (PaymentStatus), NOT NULL | References PaymentStatus table     |
| TransactionID | VARCHAR(100)  | UNIQUE                                | External transaction reference     |
| Notes         | TEXT          |                                       | Additional payment notes           |
| CreatedAt     | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP             | Record creation timestamp          |

**TABLE 13: Payout**

| Attribute  | Data Type     | Constraints                     | Description                       |
|------------|---------------|---------------------------------|-----------------------------------|
| PayoutID   | INT           | PRIMARY KEY, AUTO_INCREMENT     | Unique identifier for each payout |
| HostID     | INT           | FOREIGN KEY (User), NOT NULL    | References User table (host)      |
| BookingID  | INT           | FOREIGN KEY (Booking), NOT NULL | References Booking table          |
| Amount     | DECIMAL(10,2) | NOT NULL, CHECK > 0             | Amount paid to host               |
| PayoutDate | DATE          | NOT NULL                        | Date payout was processed         |
| Status     | VARCHAR(50)   | DEFAULT 'Pending'               | Status of payout                  |

| Attribute      | Data Type    | Constraints               | Description                       |
|----------------|--------------|---------------------------|-----------------------------------|
| TransactionRef | VARCHAR(100) |                           | Reference to external transaction |
| CreatedAt      | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP | Record creation timestamp         |

**TABLE 14: Review (Recursive)**

| Attribute   | Data Type | Constraints                             | Description                        |
|-------------|-----------|-----------------------------------------|------------------------------------|
| ReviewID    | INT       | PRIMARY KEY, AUTO_INCREMENT             | Unique identifier for each review  |
| BookingID   | INT       | FOREIGN KEY (Booking), NOT NULL, UNIQUE | References Booking table           |
| ReviewerID  | INT       | FOREIGN KEY (User), NOT NULL            | References User table (reviewer)   |
| RevieweeID  | INT       | FOREIGN KEY (User), NOT NULL            | References User table (reviewee)   |
| Rating      | INT       | NOT NULL, CHECK 1-5                     | Rating from 1-5 stars              |
| Comment     | TEXT      |                                         | Review comment                     |
| Response    | TEXT      |                                         | Host response to review            |
| ReviewDate  | DATE      | DEFAULT CURRENT_DATE                    | Date review was submitted          |
| IsPublished | BOOLEAN   | DEFAULT TRUE                            | Whether review is publicly visible |
| CreatedAt   | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP               | Record creation timestamp          |
| UpdatedAt   | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP ON UPDATE     | Last update timestamp              |

**TABLE 15: AvailabilityCalendar**

| Attribute    | Data Type | Constraints                      | Description                                |
|--------------|-----------|----------------------------------|--------------------------------------------|
| CalendarID   | INT       | PRIMARY KEY, AUTO_INCREMENT      | Unique identifier for calendar entry       |
| PropertyID   | INT       | FOREIGN KEY (Property), NOT NULL | References Property table                  |
| CalendarDate | DATE      | NOT NULL                         | Specific date for availability             |
| IsAvailable  | BOOLEAN   | DEFAULT TRUE                     | Whether property is available on this date |

| Attribute   | Data Type     | Constraints               | Description                           |
|-------------|---------------|---------------------------|---------------------------------------|
| CustomPrice | DECIMAL(10,2) |                           | Override standard price for this date |
| CreatedAt   | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP | Record creation timestamp             |

**TABLE 16: Message (Recursive)**

| Attribute   | Data Type | Constraints                  | Description                         |
|-------------|-----------|------------------------------|-------------------------------------|
| MessageID   | INT       | PRIMARY KEY, AUTO_INCREMENT  | Unique identifier for each message  |
| SenderID    | INT       | FOREIGN KEY (User), NOT NULL | References User table (sender)      |
| ReceiverID  | INT       | FOREIGN KEY (User), NOT NULL | References User table (receiver)    |
| BookingID   | INT       | FOREIGN KEY (Booking)        | References Booking table (nullable) |
| MessageText | TEXT      | NOT NULL                     | Content of the message              |
| SentDate    | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP    | When message was sent               |
| ReadDate    | TIMESTAMP | NULL                         | When message was read               |
| IsRead      | BOOLEAN   | DEFAULT FALSE                | Whether message has been read       |

**TABLE 17: Dispute**

| Attribute   | Data Type    | Constraints                     | Description                                |
|-------------|--------------|---------------------------------|--------------------------------------------|
| DisputeID   | INT          | PRIMARY KEY, AUTO_INCREMENT     | Unique identifier for each dispute         |
| BookingID   | INT          | FOREIGN KEY (Booking), NOT NULL | References Booking table                   |
| RaisedByID  | INT          | FOREIGN KEY (User), NOT NULL    | References User table (who raised dispute) |
| Title       | VARCHAR(200) | NOT NULL                        | Short title of the dispute                 |
| Description | TEXT         | NOT NULL                        | Detailed description of the dispute        |
| Status      | VARCHAR(50)  | DEFAULT 'Open'                  | Status (Open, In Progress, Resolved)       |
| Resolution  | TEXT         |                                 | How dispute was resolved                   |
| CreatedDate | DATE         | DEFAULT CURRENT_DATE            | Date dispute was created                   |

| Attribute    | Data Type | Constraints                            | Description               |
|--------------|-----------|----------------------------------------|---------------------------|
| ResolvedDate | DATE      |                                        | Date dispute was resolved |
| CreatedAt    | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP              | Record creation timestamp |
| UpdatedAt    | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP<br>ON UPDATE | Last update timestamp     |

**TABLE 18: AuditLog**

| Attribute  | Data Type    | Constraints                              | Description                                  |
|------------|--------------|------------------------------------------|----------------------------------------------|
| LogID      | INT          | PRIMARY KEY,<br>AUTO_INCREMENT           | Unique identifier for each log entry         |
| UserID     | INT          | FOREIGN KEY (User) ON DELETE<br>SET NULL | References User table<br>(nullable)          |
| Action     | VARCHAR(255) | NOT NULL                                 | Action performed (CREATE,<br>UPDATE, DELETE) |
| EntityType | VARCHAR(50)  |                                          | Type of entity affected                      |
| EntityID   | INT          |                                          | ID of the affected entity                    |
| Details    | JSON         |                                          | Additional details in JSON<br>format         |
| IPAddress  | VARCHAR(45)  |                                          | IP address of user                           |
| Timestamp  | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP                | When action occurred                         |

**TABLE 19: SocialLink**

| Attribute     | Data Type    | Constraints                     | Description                                   |
|---------------|--------------|---------------------------------|-----------------------------------------------|
| LinkID        | INT          | PRIMARY KEY,<br>AUTO_INCREMENT  | Unique identifier for each<br>social link     |
| UserID        | INT          | FOREIGN KEY (User), NOT<br>NULL | References User table                         |
| Platform      | VARCHAR(50)  | NOT NULL                        | Social platform (Facebook,<br>LinkedIn, etc.) |
| ProfileURL    | VARCHAR(255) | NOT NULL                        | URL to user's profile on<br>platform          |
| ConnectedDate | DATE         | DEFAULT CURRENT_DATE            | When social account was<br>connected          |
| IsVerified    | BOOLEAN      | DEFAULT FALSE                   | Whether social link is verified               |

| Attribute | Data Type | Constraints                  | Description               |
|-----------|-----------|------------------------------|---------------------------|
| CreatedAt | TIMESTAMP | DEFAULT<br>CURRENT_TIMESTAMP | Record creation timestamp |

**TABLE 20: HostReport**

| Attribute     | Data Type     | Constraints                    | Description                       |
|---------------|---------------|--------------------------------|-----------------------------------|
| ReportID      | INT           | PRIMARY KEY,<br>AUTO_INCREMENT | Unique identifier for each report |
| HostID        | INT           | FOREIGN KEY (User), NOT NULL   | References User table (host)      |
| PeriodStart   | DATE          | NOT NULL                       | Start date of reporting period    |
| PeriodEnd     | DATE          | NOT NULL                       | End date of reporting period      |
| TotalBookings | INT           | DEFAULT 0                      | Number of bookings in period      |
| TotalNights   | INT           | DEFAULT 0                      | Total nights booked               |
| TotalRevenue  | DECIMAL(12,2) | DEFAULT 0                      | Total revenue from bookings       |
| AvgRating     | DECIMAL(3,2)  |                                | Average rating received           |
| CreatedAt     | TIMESTAMP     | DEFAULT CURRENT_TIMESTAMP      | Record creation timestamp         |

**TABLE 21: CancellationPolicy**

| Attribute         | Data Type   | Constraints                              | Description                       |
|-------------------|-------------|------------------------------------------|-----------------------------------|
| PolicyID          | INT         | PRIMARY KEY, AUTO_INCREMENT              | Unique identifier for each policy |
| PropertyID        | INT         | FOREIGN KEY (Property), NOT NULL, UNIQUE | References Property table         |
| PolicyType        | VARCHAR(50) | NOT NULL                                 | Type (Flexible, Moderate, Strict) |
| RefundPercentage  | INT         | CHECK 0-100                              | Percentage refunded               |
| DaysBeforeCheckin | INT         |                                          | Days before check-in for refund   |
| Description       | TEXT        |                                          | Detailed policy description       |
| CreatedAt         | TIMESTAMP   | DEFAULT CURRENT_TIMESTAMP                | Record creation timestamp         |

**TABLE 22: Verification**

| Attribute      | Data Type    | Constraints                           | Description                                |
|----------------|--------------|---------------------------------------|--------------------------------------------|
| VerificationID | INT          | PRIMARY KEY, AUTO_INCREMENT           | Unique identifier for each verification    |
| UserID         | INT          | FOREIGN KEY (User), NOT NULL          | References User table                      |
| DocumentType   | VARCHAR(50)  | NOT NULL                              | Type of document (Passport, ID Card, etc.) |
| DocumentURL    | VARCHAR(255) | NOT NULL                              | URL to stored document                     |
| Status         | VARCHAR(50)  | DEFAULT 'Pending'                     | Status (Pending, Verified, Rejected)       |
| VerifiedBy     | INT          | FOREIGN KEY (User) ON DELETE SET NULL | Admin who verified                         |
| VerifiedDate   | DATE         |                                       | When verification occurred                 |
| Notes          | TEXT         |                                       | Additional notes                           |
| CreatedAt      | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP             | Record creation timestamp                  |
| UpdatedAt      | TIMESTAMP    | DEFAULT CURRENT_TIMESTAMP ON UPDATE   | Last update timestamp                      |

**QUICK REFERENCE: Key Constraints Summary**

| Constraint Type | Applied To                 | Purpose                          |
|-----------------|----------------------------|----------------------------------|
| PRIMARY KEY     | All tables (ID columns)    | Unique identification of records |
| FOREIGN KEY     | All relationship columns   | Maintain referential integrity   |
| UNIQUE          | Email, TransactionID, etc. | Prevent duplicate values         |
| CHECK           | Rating (1-5), Price > 0    | Validate data ranges             |
| NOT NULL        | Required fields            | Ensure mandatory data            |
| DEFAULT         | CreatedAt, Status fields   | Set default values               |